LESSON 6.9
EXPLORING MULTI-STEP PROPORTION PROBLEMS

LESSON INTRODUCTION

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

LEARNING TARGETS

- I can solve multi-step problems involving proportions.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- What is your approach to problem-solving?
- How can you use proportional relationships to solve a multi-step ratio problem?

LESSON NARRATIVE

This lesson extends students' proportional thinking skills to include solving multi-step ratio problems involving proportions. The 6.9.2 Guided Instruction includes exploring strategies for thinking through multiple steps in a problem, including questioning based on the given information. It then explores different strategies (unit rates vs. proportions) for solving the same multi-step problem. Students practice additional problems with their partner and compare strategies with other students before reflecting on their areas of growth in proportional thinking.

COMPONENT SUMMARY

6.9.1 WARM-UP *(~5 MIN.)*

Video for interpreting and applying ratios using a baking recipe

6.9.2 GUIDED INSTRUCTION *(10 MIN.)*

Exploring multi-step problem-solving approaches in context

6.9.3 COLLABORATION *(10 MIN.)*

Analyze the work of others in using proportional reasoning to solve a multi-step problem

6.9.4 TRY IT *(10 MIN.)*

Partners work together to solve multi-step problems on their own

6.9.5 COLLABORATION *(5-10 MIN.)*

Find quads to compare approaches and answers from 6.9.4 Try It problems

6.9.6 WRAP-UP *(~5 MIN.)*

Respond to two reflection questions about their learning in today's lesson

RESOURCES

GLOSSARY

Key Terms:

- Proportional relationship

Additional Terms:

- N/A

MATERIALS

- Video for 6.9.1 Warm-Up

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may focus only on the values in the problem and ignore the context of the problem that provides insight into making sense of what to do to solve the problem.

THINGS TO CONSIDER

- Utilize instructional time and intervention for supporting process over product. Notice the activities intentionally frame the students' approaches to problem-solving rather than their final answers. Allow this approach to inform your instructional decision-making.
- Consider providing students the opportunity to first read a context and summarize their understanding of what is happening in the context with a partner prior to answering any questions. This allows the opportunity for them to make sense of the context before trying to answer any questions.

LITERACY AND LANGUAGE ROUTINES

Algebra Talk

Consider providing an opportunity for synthesis between 6.9.3 Collaboration and 6.9.4 Try It. Monitor for student responses during 6.9.3 Collaboration for misconceptions to surface whole group before students practice in 6.9.4 Try It. Use talk moves for the Algebra Talk listed with the activity to facilitate precise mathematical language during the conversation.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

Consider a Think-Aloud and similar strategies in modeling the approach to problem-solving situations involving multiple steps. Students often need more support in knowing how to start problems and when they have the final answer rather than the calculations, so ensure your instructional time reflects the gaps and needs of your students.

DIFFERENTIATE INSTRUCTION

Mark-Up Text can be an important scaffolding strategy that contributes both to fluency by increasing efficiency as well as effortful learning as described in MA.K12.MTR.1.1. In 6.9.3 Collaboration, questions 1 and 2 ask the students to draw an asterisk and a smiley face next to specific information in the work shown that shows proportional thinking. To help students organize their thinking, instruct them to use these kinds of labels to help them organize their thinking as they create proportions in the future, particularly as problem-solving continues in Lessons 10-11. It would also be helpful to model this when thinking aloud through 6.9.2 Guided Instruction.

6.9.1 WARM-UP: BAKING WITH RATIOS

QUESTIONING

Consider project-based learning ideas for students who enjoy cooking at home. Students can make their own video and submit as an end-of-unit review or for homework as they continue problem-solving in the next two lessons.

6.9.1 Warm-Up: Baking with Ratios

Chef and author Michael Ruhlman makes baking easier by using simple mathematical ratios. He uses the following chart where each number represents the number of parts of that ingredient needed for the base recipe.

	Flour	Liquid	Egg	Butter	Sugar
Bread	5	3			
Quick Bread Muffins	2	2	1	1	1 *(if sweet)*
Cookies	3			2	1

1. Michael Ruhlman believes, "If you know ratios, it's like knowing a thousand recipes." What do you think he meant by this?
   Answers will vary. The basic ratios of ingredients are the same for each type of recipe. Like ratios, if you have the correct proportion for each component, you have many ways to make or change the recipe.

2. The ingredients for a homemade bread recipe are shown to the right. Do the ratios in the recipe match the ratios Michael recommended for bread? Explain your reasoning.

   - ☐ 2 cups warm water
   - ☐ $\frac{1}{2}$ cup white sugar
   - ☐ $1\frac{1}{2}$ tablespoons yeast
   - ☐ $1\frac{1}{2}$ teaspoons salt
   - ☐ $\frac{1}{4}$ cup vegetable oil
   - ☐ 5 – 6 cups flour

   Bread = flour : liquid

   5 : 3

   5 – 6 *cups*

   water & veg. oil

   $2 + \frac{1}{4} = 2.25$ *cups*

   No…there is not as much liquid in this recipe.

6.9.2 Guided Instruction: Thinking Through Multiple Steps

Solving real-world problems often involves multiple steps. To answer one question, a problem solver may need to ask and answer other questions using the information given in the problem. Consider the situation below.

Tahj needs to paint the four exterior walls of a large rectangular barn. The length of the barn is 80 feet (ft), the width is 50 feet, and the height is 30 feet. The paint costs $\$28$ per gallon (gal), and each gallon covers 420 square feet. Tahj wants to know how much it will cost to buy enough paint to paint the barn.

1. Complete the questions that Tahj needs to answer before she can determine the cost. Then, find the answer to each question in the workspace provided.

Question	Workspace
What is the total <u>*area*</u> Tahj needs to paint, in square feet?	$(80 \times 30) \times 2 = 4800$ $(50 \times 30) \times 2 = +3000$ $7800\ ft^2$
*How many <u>*gallons*</u> of paint will be needed to cover this area?	$7800\ \cancel{ft^2} \times \frac{1\ gal}{420\ \cancel{ft^2}} = 18.57$ $\boxed{*\ 19\ gallons}$
*How much will it cost Tahj to buy enough paint to paint the barn?	$19\ \cancel{gal} \times \frac{\$28}{1\ \cancel{gal}} = \boxed{*\$532}$

2. Place a star next to any questions you answered using a proportion or proportional thinking.

3. Studying mathematics includes making connections between concepts when solving problems. In addition to proportional thinking, what other math concept(s) did you use to solve this problem? *Area*

6.9.2 GUIDED INSTRUCTION: THINKING THROUGH MULTIPLE STEPS

TEACHER MOVES

Consider utilizing a Think Aloud to model the math thinking and reasoning that best positions students as problem-solvers with proportions. Include modeling how to mark the text to indicate important information during the process.

6.9.3 COLLABORATION: THINKING THROUGH MULTIPLE STEPS

DIFFERENTIATE INSTRUCTION

Invite students to mark up the problem as they read or listen to help them easily identify the important information.

ENRICHMENT

What information from the text needs to be used in each incomplete portion of the students' work?

6.9.3 Collaboration: Thinking Through Multiple Steps

Sean lives in Florida and frequently travels for work. Each week, he is allotted 500 U.S. dollars for business-related expenses while he is out of town. He is going to Canada this week, so he went to the bank to exchange 500 U.S. dollars for Canadian dollars (CDN) at a rate of \$1.32 CDN per \$1 U.S. On the way home from the bank, Sean's boss called to say that he has to go to Mexico City now instead. Sean went back to the bank to exchange his Canadian dollars for Mexican pesos at a rate of 16.21 pesos per \$1 CDN.

Two students used different strategies to determine how many pesos Sean has for his business trip. Their partial work is shown. Work with your partner to complete each student's work.

Maurice's Strategy Using Unit Rates	* $\$500\ \cancel{US} \cdot \frac{\$1.32\ \cancel{CDN}}{\$1\ \cancel{US}} \cdot \frac{16.21\ pesos}{\$1\ \cancel{CDN}} =$ $10{,}698.6\ pesos$
Kiara's Strategy Using Proportions	$\frac{\$1.32\ CDN}{\$1\ US} = \frac{x}{\$500\ US}$ $\frac{\$1.32\ CDN}{\$1\ US} \cdot 500 = \frac{x}{\$\cancel{500}_{1}\ US} \cdot \cancel{500}^{1}$ * $\$660\ CDN = x$ $\cancel{16.21}\ m \times \frac{\$1\ CDN}{\cancel{16.21}\ pesos} = \frac{\$\ 660\ CDN}{\cancel{m}\ pesos} \times 16.21\ \cancel{m}$ $m = 10{,}698.6\ pesos$

1. Draw an asterisk (*) next to the part of each student's work that represents Sean converting his U.S. dollars to Canadian dollars.

2. Draw a smiley face (☺) next to the part of each student's work that was used to determine the number of pesos Sean has for his business trip.

3. Both students used proportional reasoning to solve the problem in multiple steps. Which method do you prefer? Explain.

   Answer will vary. Either preference is fine. Maurice's strategy involves less steps. Kiara's strategy involves more detailed steps.

6.9.3 COLLABORATION: THINKING THROUGH MULTIPLE STEPS (CONTINUED)

TEACHER MOVES

Algebra Talk

As students share their answers for question 3, prompt other students to respond with talk moves like the following: • "Who can restate ___'s reasoning in a different way?"

- "Did anyone have the same strategy but would explain it differently?"
- "Did anyone solve the equation in a different way?"
- "Does anyone want to add on to _____'s strategy?"
- "Do you agree or disagree? Why?"

6.9.4 TRY IT: SOLVING MULTI-STEP PROBLEMS WITH PROPORTIONS

TEACHER MOVES

Teacher Note:

Notice that in 6.9.5 Collaboration, students will discuss and compare strategies used in this component.

ENRICHMENT

- What is the given ratio?
- Which part of the ratio does 15 minutes “match” with?
- What information are you comparing your answer to in order to answer the question?

6.9.4 Try It: Solving Multi-Step Problems with Proportions

1. Reina is making s'mores kits with a handmade thank-you card as party favors for her upcoming birthday party. She wants to make 9 kits, one for each party guest. It takes her about 15 minutes to assemble each kit and she doesn't have much time.

   a. Work with your partner to determine the question(s) that should be answered if Reina is worried about having enough time to make 9 kits.
      - *How long will it take her to make all the kits?*
      - *How much time does she have?*

   b. Determine if Reina will be able to make all of the kits in 2 hours. Show your work.

   $9\ \cancel{kits} \times \frac{15\ \cancel{min}}{1\ \cancel{kit}} \times \frac{1\ hr.}{60\ \cancel{min.}} = 2.25\ hours$

   She will not be able to make all the kits unless she works a little faster.

2. A store has two different brands of laundry detergent. Laundrette can do 60 loads of laundry and costs \$10.25. Peroxy does 24 loads of laundry and costs \$4.56.

   a. Complete the statement.

   If Claudia wants to compare which detergent will cost her less per load of laundry, she should find a

   ○ proportion
   ○ unit rate

   for each brand.

b. Which laundry detergent costs less per load?
Laundrette

$\frac{\$10.25}{60\ loads} \approx \frac{\$0.17}{1\ load}$

Proxy

$\frac{\$4.56}{24\ loads} = \frac{\$0.19}{1\ load}$

3. Denise makes trail mix by combining 3 parts granola, $1\frac{1}{2}$ parts dried fruit, and $\frac{1}{2}$ part chocolate. She plans to make 20 cups of trail mix for her family's upcoming camping trip. How much of each ingredient does she need?

$3 + 1\frac{1}{2} + \frac{1}{2} = 5\ parts \xrightarrow[\times 4]{} 20\ cups$

granola: $3 \times 4 = 12$ cups
dried fruit: $1\frac{1}{2} \times 4 = 6$ cups
chocolate: $\frac{1}{2} \times 4 = 2$ cups

6.9.4 TRY IT: SOLVING MULTI-STEP PROBLEMS WITH PROPORTIONS (CONTINUED)

ENRICHMENT

Did you use a proportion or a unit rate to solve question 2? Why did you make that decision?

ENRICHMENT

Did you use a proportion or a unit rate to solve question 3? Why did you make that decision?

6.9.5 COLLABORATION: COMPARING THINKING

ENRICHMENT

To support the discussion between groups, encourage the groups to focus on questions about how they got their answer instead of whether or not they got the same answer. Questions you pose might include:

- Which questions do your steps answer?
- Which ratio did you start with and why?
- Which operations did you use and why?
- How many steps did you need?
- What unit rates did you use?

DIFFERENTIATE INSTRUCTION

For quads that have the same work shown as their partner pair, ask them to consider whether their work used a unit rate or a proportion and challenge them to see if they could solve it using the other strategy.

6.9.5 Collaboration: Comparing Thinking

With your partner, find another pair of partners to make a group of four.

1. Discuss and compare your work on the Try It problems. Verify your answers and resolve any differences. If necessary, revise your work.

6.9.6 Wrap-Up: Growth Mindset

There is power in believing you can improve. Complete each statement based on your learning today.
Answers may vary.

1. Today, I am still unsure about …

2. I will try to build my confidence by …

6.9.6 WRAP-UP: GROWTH MINDSET

DIFFERENTIATE INSTRUCTION

Consider having students go back through the lesson to place a question mark on areas that they are unsure about.